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Patent Cases in the Court of Chancery, 1714–58

SEAN BOTTOMLEY

The purpose of this article is two-fold. First, it discusses recent improvements in the cataloguing of Chancery bills and pleadings entered between 1714 and 1758, held in the C 11 series at the National Archives. This has made it much easier to locate cases by subject, and a methodology for doing this is described. Secondly, the article outlines the results of work carried out in C 11 on cases relating to patents for invention. Although there has been significant research into how other forms of intellectual property right were adjudicated in the Court of Chancery, notably copyright, patent law for this period remains obscure. The article shows that Chancery (along with the common law courts) retained the principal jurisdiction in patent law, rather than the Privy Council as was once thought.

I. PATENTING IN THE PRIVY COUNCIL

The only research on patent law before the mid-eighteenth century was undertaken by Edward Wyndham Hulme a century ago, and more recently by Christine MacLeod.¹ Hulme suggested that the Privy Council was primarily responsible for adjudicating patent law until 1753, rather than the equity or common law courts.² The Council offered remedies for both patentees seeking to enforce their rights against infringement and for those who sought to have these rights annulled. For the latter, it was possible to have a patent revoked if signed for by six privy councillors.³ For patentees,

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¹See for example, Edward Wyndham Hulme, 'On the History of Patent Law in the Seventeenth and Eighteenth Centuries', 18 *Law Quarterly Review* (1902), 280–288; Edward Wyndham Hulme, 'Privy Council Law and Practice of Letters Patent for invention from the Restoration to 1794', 33 *Law Quarterly Review* (1917), 63–75, 181–195; and Christine MacLeod, *Inventing the Industrial Revolution: The English Patent System, 1660–1800*, Cambridge, 1988, ch.4.

²There were many types of patent awarded during this period. Some were granted to confer a title, others were printing patents, granting the exclusive right to print a particular book or work. Here, unless it is explicitly stated otherwise, 'patents' exclusively denote patents for invention.

³And the relevant clause permitting revocation by the Council was retained in the text of the patent until 1902. Moureen Coulter, *Property in Ideas: The Patent System in Mid-Victorian England*, Kirksville, MO, 1990, 13.

it was possible to have suspected infringers summoned to the Council and tried for contempt of the royal grant.⁴ In either instance, cases were begun by petitioning the Council with details of the grievance, and the remedy sought.

In July 1719, for example, the Privy Council received a petition from the Company of Silk Throwsters seeking the revocation of a patent awarded to Thomas Lombe for his silk 'engines'.⁵ These engines were among the most technically sophisticated and important inventions to be patented in England in the first half of the eighteenth century. Lombe, by the use of water-powered machinery, was able to wind, spin and twist silk, mechanizing operations that had previously been performed by hand (termed 'throwing').⁶ To carry out the patent, Lombe constructed an enormous manufactory at Derby of some five or six storeys, described by David Landes in *The Unbound Prometheus* as 'one of the wonders of the age'.⁷ This was not Lombe's own invention, however. Rather these engines had been developed in Italy over the course of the seventeenth century. In 1715 Lombe had tasked his half-brother (John) with travelling to Italy and 'penetrating the secrets' of the water-driven silk-throwing factories, a crime punishable in Sardinian law by death.⁸ Two years later, after much expense and hazard, his brother returned with the technology, and in 1718 a patent was obtained. Lombe's enterprise, however, prompted opposition from the Company of Silk Throwsters, which recognized the threat his engines posed to their traditional handicraft. In its petition to the Council, the Company asserted that the engines were not 'a new invention' in England (a transparently fraudulent claim) and that the 'said patent interferes with the Charter granted to the said company', excising them of trade.⁹

Petitions relating to patents were normally referred to a committee of the Privy Council (as in Lombe's case) or to one of the law officers. Although the executive authority of the Council was in decline, it remained a body of some importance. Many of the pre-eminent political and judicial figures of the day were members and they often attended committees appointed to hear patent disputes.¹⁰ For example, the committee which heard the Throwsters' petition was composed of the lord chancellor (Lord Macclesfield), the lord president of the Council (Viscount Townshend, the Whig statesman), the earl of Westmorland, the earl of Sutherland,

⁴In the case brought by Howard and Watson against the Company of Painters and Stainers for the infringement of their patent in 1669, King Charles II confirmed the validity of the patent, and decided that subsequent infringements would be prosecuted for contempt of his order, 'as a disturber of the public good and welfare of this Kingdom'. Hulme, 'Privy Council Law', 68–69.

⁵The National Archives: Public Record Office (PRO), PC 2/86, fo.294.

⁶William Chaloner, *People and Industries*, London, 1963, 9.

⁷David Landes, *The Unbound Prometheus*, Cambridge, 1969, 81.

⁸Chaloner, *People and Industries*, 12.

⁹PRO, PC 2/86, fo.294. Craft guilds were frequently opposed to patents and sometimes petitioned the Privy Council to have them revoked. This could be interpreted as evidence of technological conservatism on the part of the guilds, although Stephan Epstein argued that they were primarily concerned with protecting their collective knowledge from appropriation by individual guild members and competitors. Stephan Epstein, 'Property Rights to Technical Knowledge in Pre-modern Europe, 1300–1800', 94 *The American Economic Review* (2004), 384. In Lombe's case, however, where nothing akin to these engines had been previously used in England, Epstein's argument cannot hold and the Company's opposition was most likely an attempt to maintain its privileged position.

¹⁰Peter Jupp, *The Governing of Britain, 1688–1848*, Abingdon, 2006, 23.

the chief justice of the Common Pleas (Lord King) and Vice-Chamberlain Coke.¹¹ Although the Privy Council registers only provide the final report, it is likely that proceedings were attended by legal counsel for both sides – certainly the law officers were when conducting similar hearings.¹²

The committee rejected the petition of the Throwsters because they were unable ‘to prove that any of the said engines were ever before made or used in this Kingdom’.¹³ The petition was also rejected because the Throwsters had failed to prove their second substantive point against the patent: that they were being excised ‘of trade’, and that Lombe’s engines were causing unemployment.¹⁴ Although unsuccessful, the Throwsters had been well advised to make this claim; wider considerations of the ‘public good’ permeated decision making in patent cases before the Council. One of the positive reasons the committee gave for upholding the legality of Lombe’s patent was his ‘particular sort of silk appearing to be a usefull manufacture which hath hitherto been imported from Italy and now by the said Engines is made in great perfection in England’.¹⁵ This concern with the public interest (largely defined by economic/mercantilist priorities) was entirely consistent with the Statute of Monopolies (1624) which held that patents could not be ‘generally inconvenient’.¹⁶ Although vaguely worded, Sir Edward Coke elaborated upon this condition in his authoritative *Institutes of the Laws of England*, first published between 1628 and 1644.¹⁷ Coke had not only chaired the committee that passed the bill for the Statute, but he had also been plaintiff’s counsel in *Darcy v Allen* (1602), one of the first reported cases at common law on patents for manufactures.¹⁸ Notably, Coke considered a patent to be ‘inconvenient’ if it caused unemployment, citing a 1553 statute of Edward VI where ‘it was ordained that bonnets and caps should be thickened and fulled by the strength of men, and not in a fulling mill, for it was holden inconvenient to turn so many men to idlenesse’.¹⁹

This indicates how malleable and fluid considerations of the public good could be. Although Lombe had managed to maintain his patent, strictly speaking this contravened the Statute of Monopolies and Coke’s *Institutes*. Certainly the maintenance of this requirement was not conducive to a settled system of property rights. To give

¹¹PRO, PC 2/87, fo.144.

¹²See for example *Hamblin’s Case* (1731) PRO, PC 1/5/4.

¹³PRO, PC 2/87, fo.144.

¹⁴Later, when opposing Lombe’s petition to Parliament to have his patent term extended, the Company revised this argument, arguing that they should be allowed to use the engines themselves, asserting that ‘by [Lombe’s] case printed it appears that the said engines employ’d about 300 persons; which, upon a reasonable supposition, is not the tenth part of the number which the Silk Throwers could employ, if they were at Liberty to make and use the said Engines’. Parliamentary Papers, ‘The Case of the Silk Throwers Company’, 1732 (House of Commons).

¹⁵*Ibid.*

¹⁶21 Jac. I, c.3.

¹⁷Edward Coke, *The Institutes of the Laws of England*, 4 vols., 4th ed., London, 1669, vol.3, 181–185.

¹⁸Jacob Corré, ‘The Argument, Decision, and Reports of *Darcy v Allen*’, 45 *Emory Law Journal* (1996), 1262. *Darcy v Allen* concerned a patent privilege for the sole importation, manufacture and sale of playing cards: a monopoly. Although the right of the crown to make such grants was contested, defence counsel recognized that there was one area where no common law restriction on the Crown’s authority to award trade privileges existed, the patent for invention. *Ibid.*, 1304.

¹⁹Coke, *The Institutes*, 184. The Statute is 7 Edw. VI, c.8.

another example, in 1721 two separate parties applied for patents for similar inventions relating to the smelting of iron at almost the same time. The matter was referred to a committee of the Privy Council as well as the Attorney-General Philip Yorke (later Lord Hardwicke). In his report, Yorke observed that although Mr Forrest had ‘the preference to your Majesty’s favour in respect to his priority of application’, on ‘the point principally referred I was of opinion that the petitioners [Brown and Wright] said invention might be most serviceable to the Publick’.²⁰ This was largely because Brown and Wright had already ‘built several works for that purpose and it is likely to prove so beneficial to the Nation in General by preserving the Wood and Timber which is now consumed in such large quantities by the present method’.²¹ Unfortunately for Brown and Wright, the Council ignored Yorke’s recommendation and resolved the dispute in a summary manner. By deciding that both parties had presented mere ‘improvements’, which were not covered by the provisions of the Statute of Monopolies (which referred only to ‘manufactures’), the Council found that neither party was entitled to a patent, and this decision was later confirmed by an order of the king in Council.²² Although the Committee’s report is terse and the logic is not explained, they likely thought that the public good would best be served by opening up the technology belonging to both parties for free use.

It appears then that the rights of inventors and patentees were of secondary importance to the Privy Council when compared to wider considerations of the public good. The risk and uncertainty of the patent grant was exacerbated by the fact that decisions in the Council were not perceived to be binding upon the adjudication of subsequent disputes. As MacLeod recognizes, it could only be ‘when the Privy Council relinquished jurisdiction, that a case law on patents [could] develop’.²³ It is generally thought that this happened in 1753, prompted by an unusual chain of events. In the year before, a Walter Baker had petitioned the Council, requesting that the patent belonging to the chemist Dr James be annulled. The patent covered a lucrative medicine known as ‘Dr James Fever Powders’, which reputedly cured a host of ailments such as gout, fever, rheumatism and scurvy. By 1764 1.6 million doses had been sold.²⁴ Baker claimed in his petition that the powder had been invented not by James, but a Baron Schwanberg, and in support Baker produced forty-three affidavits, whereas it appears that James was only able to produce one – his own.²⁵ The Council referred the petition to the law officers, who advised, despite the number of affidavits obtained by Baker, that none of the evidence was ‘pertinent’, and that James should be allowed to keep his patent.²⁶ Undaunted, Baker prosecuted James for perjury his

²⁰PRO, PC 1/3/102, fo.2.

²¹Ibid.

²²More precisely: ‘it not being made appear that either of the petitioners have found out any new invention but only made improvements in the manner of melting iron ore. Their Lordships are therefore humbly of opinion that neither party hath any title to his Majesties Letters Patent’. For a copy of the king’s order, see PRO, SP35/63, fo.90 (1).

²³MacLeod, *Inventing the Industrial Revolution*, 61.

²⁴Lionel Bently, ‘Patents and Trade Secrets in England: The Case of *Newbery v James* (1817)’, mimeo (2012), 2.

²⁵Walter Baker, *The Affidavits and Proceedings of Walter Baker*, London, 1754.

²⁶PRO, PC 2/103, fo.321.

affidavit to the Privy Council, and petitioned the Council for its clerk to attend King's Bench with James's original affidavit.²⁷ While the law officers decided on the merits of this request, 'the petitioner Mr Baker clandestinely got away from the Attorney-General's Office several material affidavits filed there on occasion of the original petition' and printed them as a pamphlet. As a result of his malfeasance, Baker was refused permission for the clerk of the Council to attend and there were no further proceedings. Hulme claimed that it was this debacle that prompted the Privy Council to surrender its jurisdiction.²⁸

If Hulme is correct, and it was not until 1753 that the Privy Council surrendered its jurisdiction in patent cases, this would mean that the security of patent rights in the first half of the eighteenth century was very weak. This would have undermined their pecuniary value, in turn undermining the incentive to engage in inventive activities theoretically provided by patents.²⁹ However, as MacLeod observes, 'in the light of earlier controversies, it was a curiously unremarked transference of authority', and in the absence of direct evidence, Hulme's suggestion is speculative.³⁰ Furthermore, the case-load of the Privy Council had been in decline long before 1753; a more plausible dividing line is 1688, the year of the Glorious Revolution. Although the political and economic consequences of the Glorious Revolution are still hotly contested, it did signify the supersession of the Privy Council, by Parliament, as the principal regulator of the national economy.³¹ For example, between 1660 and 1688, the Privy Council issued an average of four Royal Proclamations concerning the economy every year, whereas between 1689 and 1714, the average fell to less than one Proclamation a year.³² There was a concomitant withdrawal from patent affairs as well. With the

²⁷Ibid., fo.422.

²⁸Hulme, 'Privy Council Law', 194.

²⁹Kenneth Arrow, 'Economic Welfare and the Allocation of Resources for Invention', in National Bureau of Economic Research (NBER), *The Rate and Direction of Inventive Activity: Economic and Social Factors*, Princeton, NJ, 1962, 609–626.

³⁰MacLeod, *Inventing the Industrial Revolution*, 59. Moreover, as Hulme was aware, there was one important dispute resolved by the Privy Council after 1753. In 1774 John Wilkinson took out a patent for his cannon-boring machine which would be later used by James Watt to bore the cylinders for his steam engines. A year later, in 1775, the Board of Ordnance submitted a petition to the Privy Council claiming that 'Messers James & Clark who are under contract with us beginning to work according to that method were prevented from proceeding by notice from John Wilkinson that he had obtained his Majesty's Letters Patent for the sole making and vending such sort of Iron Ordnance'. The Board argued that the matter was 'of such great consequence to His Majesty's Service that [we] must entreat your Lordships to lay the same before His Majesty in Council'. PRO, PC 1/10/75. The affair was referred to the law officers, who were reluctant 'to make any report . . . in expectation that the matter would be tried at action' – and they only did so after a four year interval in 1779. They noted that Wilkinson had filed a bill in Chancery 'and obtained the common injunction to restrain them from working', but that he had then failed to pursue the case to common law. Concluding that Wilkinson had only engaged in a delaying action to alarm competitors, they recommended that the patent should be annulled. Their forbearance is especially noteworthy as the American Revolutionary War had just entered its fourth year. PRO, PC 1/11/150, fo.7. The Judicial Committee concurred with the report, stating that 'the said patent if permitted to remain in force will not only be prejudicial to his majesty's subjects but very detrimental to the public service as the providing of iron ordnance will thereby become a monopoly', and the patent was annulled. PRO, PC 1/11/150, fo.2.

³¹For an overview of this debate see Steven Pincus and James Robinson, 'What Really Happened during the Glorious Revolution?', NBER Working paper no.17206 (2011), 2–5.

³²William Pettigrew, 'Debating the 'National Interest': Some Underappreciated Connections between Constitutional Change and National Economic Growth in England, 1660–1720', mimeo (2009), 7.

revival of the patent system in 1660, heralded by the Restoration, to the Glorious Revolution (a time-span of twenty-nine years), there were twenty cases relating directly to patents of invention and petitions for patents in the Privy Council. In this period, 134 patents were awarded. By contrast, from 1689 to 1753 (a period of sixty-five years and when 415 patents were awarded), there were fourteen cases in the Privy Council, just two of which occurred after 1730.³³

For inventors taking legal action to enforce their patents, the Council had been an irrelevancy long before 1753. This is significant for two reasons. First, as has been discussed, the legal priorities of the Privy Council, if extensively enforced, would have undermined the security and value of patents. Secondly, there is the question of the legal status of patents for invention. Were patents ‘merely’ a royal privilege that could be arbitrarily revoked by six of the King’s Privy Councillors? Or were they subject to consistent forms of legal scrutiny in the common law and equity courts, perhaps in a manner analogous to ‘property rights’ and with the remedies that this would entail? This speaks directly to how the English legal system was beginning to treat not just patents but other forms of intellectual property right.

II. PATENTING IN THE COURT OF CHANCERY? A METHODOLOGY

In essence, a patent represents the right to exclude others from employing the protected subject matter. However, without the means of enforcing this exclusion against infringements, a patent will, in effect, be worthless. If this were so, it would be difficult to account for the inventors (relatively few though they were) who went to the trouble of obtaining patents between 1688 and 1753, an apparent black hole in the enforceability of patent rights.

Although MacLeod discusses the equity Courts of Chancery and Exchequer as an alternative jurisdiction to the Privy Council for patentees, one where they could at least obtain financial redress, she suggests that in practice they were of only marginal importance. This is attributed to the reluctance of Chancery and Exchequer to retreat from a conventional operation of equity. Justifications for equity traditionally began from the premise that a system of laws could not achieve the ends of justice in every conceivable set of circumstances. As such, it was thought necessary to provide some means by which parties could avoid the harsh strictures of common and statute law, by offering a ‘resort to natural principles’, unfettered by any binding commitment to prior practice. In his introduction to *The Accomplished Practiser in the High Court of Chancery*, published in 1741, Joseph Harrison speculated on the origins of equity:

inasmuch as positive written Laws consist of general Institutions, grounded upon that which happeneth for the most part, the utmost extent of human wisdom being incapable of foreseeing every particular thing which time and experience doth beget, there were a variety of particular cases frequently happening, wherein no proper remedy could be given by the ordinary courts...

³³The data for the quantitative work on cases in the Privy Council is based on Hulme, and corroborated with my own work in the Privy Council Registers. I found no new cases in the Privy Council Registers. Hulme, ‘Privy Council Law’.

therefore to supply the want and relax the rigour of the positive laws, recourse was had to the King.³⁴

Because positive law and prior practice did not have an informative role in subsequent decisions, MacLeod argues that Chancery (and Exchequer) did not provide adequate security to patentees, commenting that 'The legal situation was highly uncertain. The equity courts were rarely guided by precedent . . . fears of the expense involved were compounded by the arbitrariness of the courts'.³⁵ Consequently, MacLeod suggests that few patentees prosecuted infringers before 1815, and especially before 1770.³⁶ Although she does not provide figures for overall activity, MacLeod refers to only four separate cases between 1714 and 1758 (the period covered by the C 11 pleadings class in the National Archives) in the bibliography to her monograph, published in 1988.³⁷

There are, however, two good reasons for a reappraisal. First, Chancery was not so wedded to the classical notion of equity as is supposed by MacLeod. Unlike the rest of Europe, the adjudication of equity was largely confined to its own courts.³⁸ These courts, of necessity, were obliged to formulate a system of rules and forms of procedure that in time transmuted into something akin to law.³⁹ In the large majority of cases, Lord Chancellors decided not by equitable appeals to 'natural' law but in accordance with a settled pattern of equitable practice. For example, Lord Hardwicke, who was Lord Chancellor for much of the period in consideration here (1736–56), regularly refused what he recognized to be the equitable decision because he was not prepared to 'break in on an established rule, and make a precedent of bad consequence'.⁴⁰ Even before the end of the seventeenth century, David Yale has shown how during Lord Nottingham's time on the Woolsack (1675–82) precedent was regularly cited in counsels' argument and was often used in the judicial process.⁴¹

Secondly, MacLeod recognizes that 'cases in Chancery are particularly hard to unearth because of their poor indexation. It is possible, therefore, that a number have slipped the net'.⁴² Subsequent work in analogous legal areas suggests that this is a very likely possibility. Tomás Gómez-Arostegui has recently uncovered a substantial case-load in the Court of Chancery for this period relating to copyright, demonstrating that copyright holders were regularly availing themselves of

³⁴Joseph Harrison, *The Accomplished Practiser in the High Court of Chancery*, London, 1741, 5.

³⁵MacLeod, *Inventing the Industrial Revolution*, 61.

³⁶*Ibid.*, 68.

³⁷These are *Stoughton v Wilkinson* (1722), Chancery Pleadings 1714–58, C 11/46/38, National Archives; *Newsham v Ffowke* (1729), C 11/699/10; *Kay v Tattersall* (1736), C 11/2452/15 and *Kay v Butterworth* (1737), C 11/2456/21; *ibid.*, 238 and 243.

³⁸David Lieberman, *The Province of Legislation Determined, Legal Theory in Eighteenth-Century Britain*, Cambridge, 1989, 75.

³⁹Michael Lobban notes that 'by the 1770s, equity itself had largely hardened into a precedent-based system'. Michael Lobban, *The Common Law and English Jurisprudence, 1760–1850*, Oxford, 1991, 1.

⁴⁰Lieberman, *The Province of Legislation*, 82.

⁴¹Nottingham also emphasized the limits to equitable discretion that he could exercise. David Yale, *Lord Nottingham's Chancery Cases*, 2 vols., London, 1957 and 1961, vol.1, xlv–xlvi.

⁴²MacLeod, *Inventing the Industrial Revolution*, 68.

protection, specifically injunctions, in Chancery.⁴³ Some of these right holders were protected by printing patents, and it would be surprising if holders of patents for invention (or at least their barristers) failed to appreciate what this implied for the enforcement of their own rights.

However, published reports of equity cases and handbooks discussing Chancery practice are almost entirely silent on patents for invention during this period. It is therefore necessary to re-search Chancery indices. The first stage to any suit in equity was the plaintiff's bill of complaint. Although in its usual form it contained nine separate parts, in essence it described some complaint of fraud or injury and the relief sought from the Court (which was supposed to be unavailable at common law). Quantitative work on Chancery bill books reveals a pronounced decline in Chancery's case-load during the first half of the eighteenth century. In the legal year 1700/01 5,707 bills are recorded, 3,452 in 1720/21, and a low of 1,023 in 1752.⁴⁴ Furthermore, only about three-quarters of these bills actually survive.⁴⁵ This, however, still leaves an enormous number of surviving pleadings to search through; there are probably more than 100,000 in C 11 alone. Until very recently, this mass of material was poorly indexed. In general, all that was available on the National Archives computer catalogue was the case name and year, and even then a wide range of dates might be provided (e.g. '1714–1760').⁴⁶ Horwitz and Polden note that the study of civil litigation in the Court of Chancery is 'in its infancy due in part to the Court's mass of ill-indexed and complexly organized records'.⁴⁷ Admittedly, Horwitz and Polden were writing in 1996 and there has since been some important work in specific areas of Chancery litigation. Still, a coherent overview of the subject is missing.

Although research in the indices of Chancery is anything but straightforward, there are two important advantages with researching patents for invention. First, equity did not provide the means for another party to have a patent revoked.⁴⁸ Consequently, any substantive proceeding relating to patents must have been instigated by

⁴³Tomás Gómez-Arostegui, 'What History Teaches Us about Copyright Injunctions and the Inadequate Remedy-at-Law Requirement', 81 *Southern California Law Review* (2008), 1197.

⁴⁴C. Brooks, 'Interpersonal Conflict and Social Tensions: Civil Litigation in England, 1640–1830', in A.L. Brier, David Cannadine and James Rosenheim, eds., *The First Modern Society, Essays in English History in Honour of Lawrence Stone*, Cambridge, 1989, 361.

⁴⁵The 1785 bill book records 1,544 bills, but only 1,110 survive in C 12. Henry Horwitz and Patrick Polden, 'Continuity or Change in the Court of Chancery in the Seventeenth and Eighteenth Centuries?', 35 *Journal of British Studies* (1996), 29.

⁴⁶The online catalogue can be found at: <http://discovery.nationalarchives.gov.uk/SearchUI/search/advanced-search> (accessed 26 Sept. 2013).

⁴⁷Horwitz and Polden, 'Continuity or Change', 27.

⁴⁸It was possible to have a patent for invention revoked on a writ of *scire facias*, obtainable on application to the law officers. The first reported case relating to patents for invention was *Rex v Arkwright* (1785), 1 HPC 263, although an earlier example (*Rex v Jacob* (1782)) appears in the Mansfield Notebooks. James Oldham, *The Mansfield Manuscripts and the Growth of English Law in the Eighteenth Century*, 2 vols., London, 1992, vol.1, 767. In theory, however, there is no reason to suppose that patents for invention could not have been revoked by writ of *scire facias* before this time (as other types of patent were). In a 1639 proclamation, for example, Charles I declared that for all 'patents for new inventions not put in practice within three years next after the date of the said grants . . . there shall be a proceeding against the said patentees, by *Quo Warranto* or *Scire facias* to recall the said Grants and Patents'. Robert Sanderson, *Foedera, Conventiones, Literæ*, London, 1735, vol.20, 341.

the patent holder – who will be listed as the plaintiff. This has the effect of halving the number of pleadings to be searched. Secondly, all patents for invention in this period were listed by Bennet Woodcroft in the 1850s.⁴⁹ Between 1714 and 1758, there were 339 patents awarded and for each patent, Woodcroft's index provides the name(s) of the original patentee(s) and the date on which the patent was awarded. Because patents were generally awarded for a fourteen-year period, this limits the time-range within which a patent could be litigated. By searching for the surname of a patentee in the C 11 pleadings class, for the years in which their patent would have been in force, it is possible to substantially reduce the number of pleadings to search through. An example of an amenable surname is 'Scobell', from a Francis Scobell who patented a pumping engine in 1724. A search in C 11 records for between 1724 and 1738 produces eleven cases where a Scobell was one of the parties – but only five where a Scobell was a plaintiff, one of which related to Francis Scobell's patent.⁵⁰

This methodology is useful for instances where there is only limited information available on the identity of a prospective litigant, although the method is somewhat rudimentary. In particular, for common surnames it does not exclude nearly enough pleadings for physical checking to become feasible. If, for example, cases relating to a 1725 patent obtained by Thomas Smith had been searched for, this would have necessitated checking almost 800 cases where the plaintiff had the surname 'Smith', between 1725 and 1739. Even for unusual surnames such as Scobell, it still involves an element of physical checking. It is worth mentioning, though, that this methodology can also be used when searching for cases in C 12 (covering bills of pleading between 1758 and 1800) and C 13 (covering bills of pleading between 1800 and 1842).

Thanks, however, to very recent improvements to the online catalogue at the National Archives, a much more refined means of searching for cases in C 11 is possible. Now the catalogue provides the full names of all parties, a precise case year and very often the occupations and residencies of the plaintiff(s) and defendant(s).⁵¹ Although residency and occupation information are not always available in the Woodcroft index, where it is, in the case of a match in information between the plaintiff of a suit and of a patentee, there can be a fair degree of confidence that the plaintiff and the patentee were one and the same. For example, in 1748 an apothecary from Hull called Daniel Bridges obtained a patent (for purifying rape seed oil). A search for just 'Bridges' between 1748 and 1758 reveals eight cases in C 11 of a plaintiff with this surname. However, a search for 'Daniel Bridges' reveals two cases where

⁴⁹Woodcroft actually produced four indexes, each with the same population of patents, but with information that supplemented the other three, and arranged accordingly. For details, see Alessandro Nuvolari and Valentine Tartari, 'Bennet Woodcroft and the Value of English Patents, 1617–1841', 48 *Explorations in Economic History* (2011), 99. The most easily available index is Bennet Woodcroft, *Alphabetical Index of Patentees of Inventions, 1617–1852*, 2nd ed., London, 1969.

⁵⁰*Scobell v Bolitho* (1730) PRO, C 11/84/11.

⁵¹This information has been derived from Bernau's notebooks under the supervision of Liz Hore and Amanda Beran of the PRO where a microfilm copy is also available in the microfilm drawers in the Map and Large Document Reading Room. For more details on the Notebooks, and C 11 see Henry Horwitz, *Chancery Equity Records and Proceedings, 1600–1800*, London, 1988, 71–73.

the plaintiff had this full name – and in both cases the catalogue provides his occupation (apothecary) and his residency (Hull). Sure enough, these were two patent cases.⁵² Again, care should be taken with this method. It is quite possible that a patentee (or any other person of interest), will have changed their occupation and/or residency between the time when they obtained the patent (or whenever this information is available) and when they go to court. Care should also be taken with alternate spellings to surnames (for example, ‘Bridges’ could possibly appear as ‘Bridge’). In addition, many bills from 1714 to 1758 are actually held in C 12 (perhaps over 15 per cent).⁵³

When dealing specifically with patents, this method has two additional flaws. First, during this period it was possible for patentees to assign their rights in their entirety and this may not have been uncommon. Certainly a number of patents litigated in Chancery had been partially assigned. John Kay, for example, worked his flying shuttle with two partners.⁵⁴ The method used here will exclude patents that had been assigned in their entirety prior to any legal action. If a patent was assigned, then any subsequent case would not be in the name of the original patentee (as recorded by Woodcroft), but in the name of the assignee. Because there is no register of who these assignees were, a systematic trawl for cases involving assigned patents, along the lines discussed here, is impossible. Unfortunately, this means that a comprehensive list of all relevant cases will be unattainable without searching tens of thousands of pleadings. Secondly, even in cases where there is a direct match in the information available between a plaintiff and a patentee (meaning they are most likely the same person), there is no guarantee that the case will actually pertain to the patent. Indeed, this happens on a fairly regular basis. Ultimately, the only way of being sure whether a pleading is of use is by ordering and reading it in the National Archives.

Using this method, I have found forty-one bills relating directly to patent law and there are likely to be many more lurking in C 11, as yet undiscovered.⁵⁵ These bills are distributed throughout the period. The earliest case was *Darby v Bartlett* (1716), which gives no indication that this was in any way novel; the latest was *Zomer v Gapper* (1754).⁵⁶ Further, these forty-one bills involved twenty different patents, suggesting that patentees were relatively litigious. Of 339 patents awarded between 1714 and 1758, this sample shows that at least eighteen were litigated in the Court of Chancery, 5.3 per cent of the total (two were awarded before 1714). When

⁵²*Bridges v Fowler* (1750) PRO, C 11/2134/8; *Bridges v Martin* (1750) PRO, C 11/2133/3.

⁵³Horwitz, *Chancery*, 71 and 73.

⁵⁴Upon the expiration of Kay’s patent in 1747, Kay held a four-fifteenths share of the patent, a Mr Smith a three-fifteenths share, and a Mr Carter eight-fifteenths. *Kay v Mills* (1748) PRO, C 33/389, fo.450.

⁵⁵It is surprising that none of these cases was reported. In his early work on copyright injunctions between 1660 and 1800, Tomás Gómez-Arostegui found 220 cases relating to copyright and other printing rights, thirty-six of which were reported in print. Gómez-Arostegui, ‘Copyright Injunctions’, 1223. He has since discovered almost 300 cases in total. Tomás Gómez-Arostegui, Register of Copyright Infringement Suits and Actions From c.1560 to 1800. Available online at: <http://www.oldcopyrightcases.org> (accessed 9 April 2013).

⁵⁶*Darby v Bartlett* (1716) PRO, C 11/1721/15; *Zomer v Gapper* (1754) PRO, C 11/1601/39. Indeed, the first bill to be entered into Chancery after the Restoration, complaining that a patent for invention was being infringed, was in 1662. *Tomlyn v Stephens* (1662) PRO, C 5/43/113.

compared to the case-load of the Privy Council, it is clear that Chancery was the much more important jurisdiction for patent law. Also, when one considers the *enforceability* of patents, it is clear that the Council was an irrelevancy. Indeed, the Privy Council may never have been an important jurisdiction for the enforcement of patent rights, but only for those who sought their revocation. In total there are only two examples of a patentee trying to enforce their rights in the Privy Council, one in 1661 the other in 1669.⁵⁷

III. REMEDIES IN THE COURT OF CHANCERY

Of course, having established the preference of patentees for the equitable jurisdiction of the Court of Chancery, it is important to understand why – what were the remedies available and were they effective? Bills in Chancery were normally brought for matters where the plaintiff was supposed to be without remedy at common law; bills usually recited the following words ‘*Forasmuch as your orator is remediless in the premises of the common law*’, before detailing the nature of the complaint.⁵⁸ This is not to suggest that the common law had no jurisdiction in patenting before 1753 (a matter to be discussed later), but that Chancery offered remedies that were otherwise unavailable, specifically: discovery, accounts and, above all, injunctions.

Discovery was a request for information that the plaintiff claimed to be important to prosecuting the case. It was common, for example, for plaintiffs to request the names of any associates of the defendants who might be involved with the infringement of the patent. Also, if the defendant conducted their affairs in secret (which would be sensible if one was knowingly infringing a patent), it might be difficult for a patentee to obtain sufficient evidence to prove the infringement in court.⁵⁹ In *Morris v Bransom* (1776), to prove the infringement, the plaintiff had been obliged to set up a telescope on a hill overlooking the defendant’s manufactory.⁶⁰ In Chancery, however, it was possible to compel defendants to detail their methods of manufacture in their answer to the plaintiff’s bill. In *Kay v Tattersall* (1736), for example,

⁵⁷The first instance was George Tomlyn’s 1661 petition. MacLeod, *Inventing the Industrial Revolution*, 59. The second instance was the dispute involving Philip Howard and the Company of Painters and Stainers in 1669. Hulme, ‘Privy Council Law’, 68–69. Considering the controversy surrounding patents in the reign of Charles I, the Council appear to have been wary of meddling in those patents that had already been awarded. For example, in 1664 a paper entitled ‘Considerations of Lawe upon the Patent of John Garill, Wyre Drawer’ was prepared for the Council. The third ‘Consideration’ stressed ‘That by the expresse words in the Statute against Monopolies, All Monopolies and all Priviledges of sole buying, selling, working, making or using anything, shall be examined, heard, tried and determined by the Common Lawes of the Realme, and not otherwise . . . Whereas the Newness of an Invention the Countrey only can Trye, and the Legality of a Patent, the Judges only can legally determine’. Reproduced in Hulme, ‘Privy Council Law’, 66–67.

⁵⁸Giles Jacob, *The Compleat Chancery-Practiser*, London, 1730, 143 and 150.

⁵⁹Helen Gubby provides an interesting example of the potential importance of discovery to the success of legal action. In a 1791 letter to his lawyer, James Watt wrote ‘If they mean to keep the construction of the Engine secret on the hopes to prevent our being able to prove the infringement they deceive themselves for if we do not obtain proof in an easier manner we shall make application to the Lord Chancellor to enforce a discovery of their manner of working’. Quoted in Helen Gubby, *Developing a Legal Paradigm for Patents*, The Hague, 2012, 31.

⁶⁰1 HPC 187.

James Kay sought discovery on the grounds that Tattersall and his confederates worked 'in a secret and clandestine manner . . . your orators cannot find out or discover who they are. Your orators cannot come at sufficient proof whereupon to proceed against them, although it is plain and evident from the nature of their work and weaving that the same must be worked or weaved with the shuttle'.⁶¹ There followed a list of queries Kay sought information on in Tattersall's answer, such as the number of shuttles sold, at what price and to whom. Tattersall was able to sidestep some of these questions by denying he was infringing the patent, but in so doing, he was obliged to detail his own method of working and how it allegedly differed from Kay's shuttle. It follows then that Chancery provided inventors the means with which they could determine whether an infringement was taking place, who was responsible, and how they were benefited.

Secondly, to deduce the profits accruing to the defendants from the alleged infringement, plaintiffs could request accounts from the defendants. In another case involving James Kay (*Kay v Farrow* (1741)), Kay made sure to assert that he was 'was utterly remediless in the premises by the strict rules of the common law of this realm' before asking 'that they the said confederates may account with your orators for the profit thereof'.⁶² If a patentee successfully proved the infringement of their patent, it was then possible to claim for the profits made by the defendants.

Thirdly, Chancery provided patentees with injunctive relief. An injunction was a court order restraining the defendant from infringing the patent, or risk being found in contempt of court and possibly imprisoned.⁶³ This is important, and the practice and availability of injunctions is worth detailing. As has been mentioned before, patents represent the right to exclude other parties from using a protected technology. Injunctions provided the means with which patentees could directly enforce this exclusion against competitors. Unsurprisingly, every bill of pleading relating to the infringement of patents sought an injunction, and this was the case in copyright as well.⁶⁴

In cases of waste (defined as 'damage to the freehold, or inheritance of another, by felling timber, ploughing meadow, &c', normally by a tenant), Chancery would award an *ex parte* injunction to restrain the act of waste upon the bill of pleadings being submitted and an affidavit verifying the plaintiff's property right. The injunction would also be awarded if the defendant failed to respond or appear on time: within four days if they lived in London, and eight if they lived in the country.⁶⁵ In cases of copyright and patents, Chancery proceeded on a similar premise, that if the infringement of the plaintiff's right was allowed to continue, it could irreparably damage their property. In 1789, John Mitford (later Lord Redesdale) noted 'the courts of equity seem to have proceeded upon a similar principle in the very common cases of persons claiming copy-right of printed books, and of patentees of alleged

⁶¹*Kay v Tattersall* (1736) PRO, C 11/2452/15, fo.1.

⁶²*Kay v Farrow* (1741) PRO, C 11/1575/26, fo.1.

⁶³Henry Maddock, *A Treatise on the Principles and Practice of the High Court of Chancery*, 2 vols., 2nd ed., London, 1820, vol.1, 224.

⁶⁴Gómez-Arostegui, 'Copyright Injunctions', 1234.

⁶⁵William Bohun, *Cursus Cancellariæ: Or the Course of Proceedings in the High Court of Chancery*, 2nd ed., London, 1723, 440.

inventions'.⁶⁶ If defendants failed to appear within these stringent time-limits (which was uncommon before the mid-eighteenth century, at least in copyright cases), it was relatively straightforward for patentees to obtain an injunction.⁶⁷

Once awarded, an injunction could only be lifted when the defendant had entered a 'full and perfect' answer, which often took some time. In *Ribright v Hoffner* (1750), for example, the defendant entered his answer three months after the bill of complaint had been entered.⁶⁸ In his eventual answer, Hoffner complained that he had been publicly selling the patented 'perspective glasses' a year before Ribright's patent had been granted and had continued 'to sell the same until he received the injunction of this Honourable Court'.⁶⁹ Furthermore, Chancery order books (held in C 33 and which record any orders or decrees made relating to a particular case) reveal that the conditions for a 'full and perfect answer' were strictly enforced. Many defendants were obliged to obtain orders for more time to enter an answer. Similarly, many of the orders are for the referral of the answer to a Master in Chancery to decide if the answer was indeed 'full and perfect'.⁷⁰ These pre-trial proceedings could take many years. For example, the first order relating to *Kay v Mills* was in Michaelmas Term 1744⁷¹ yet the case took nearly three years to come to an oral (*inter partes*) argument before the Lord Chancellor in Easter Term 1747.⁷²

The second type of injunction, the temporary injunction, could only be obtained after such an *inter partes* argument before the Lord Chancellor. Of the forty-one initial pleadings only three came to such a hearing, *Newsham v Gray* (1737), *Kay v Mills* (1744) and *Sage v Harrison* (1749).⁷³ Some cases did not reach a hearing because, if the defendant submitted to the *ex parte* injunction immediately, it would not be worth the plaintiff's time to pursue them for what profits they had made.⁷⁴ In other instances, it can be inferred from the order books that parties reached an extra-judicial agreement; there are some cases where after years of orders, and the cause on the verge of being heard, the case disappears from the order books altogether. In *Kay v Bearcroft* (1746), for instance, the day before the hearing was scheduled, the defendant was awarded one more day, ostensibly to produce witnesses relating to the case – yet after this, there is no record of a

⁶⁶John Mitford, *A Treatise on the Pleadings in Suits in the Court of Chancery by English Bill*, 2nd ed., Dublin, 1789, 124.

⁶⁷Gómez-Arostegui, 'Copyright Injunctions', 1229. Although such affidavits were also necessary in waste, there were some (very specific) circumstances where not even affidavits were required to obtain the injunction. Bohun, *Cursus Cancellarie*, 450.

⁶⁸The bill of complaint of Ribright is dated 18 July 1750, Hoffner's first answer was entered 15 Oct. 1750. PRO, C 11/2510/12.

⁶⁹PRO, C 11/2510/12, fo.3.

⁷⁰For example, in *Kay v Smith* (1742), the answer of the defendant was referred to Master Spicer, who was to report within '4 days'. *Kay v Smith* (1742) PRO, C 33/377, fo.381.

⁷¹*Kay v Mills* (1744) PRO, C 33/383, fo.28.

⁷²*Ibid.*, fo.697.

⁷³*Newsham v Gray* (1737) PRO, C33/378, fo.190; *Kay v Mills* (1744) PRO, C33/387, fo.697–678; *Sage v Harrison* (1749) PRO, C33/394, fo.244.

⁷⁴Justice Willes in *Millar v Taylor* (1769) made this point in relation to copyright, 'Few bills against pirates of books are ever brought to a hearing. If the defendant acquiesces under the injunction, it is seldom worth a plaintiff's while to proceed for an account; the sale of the edition being stopped'. Quoted in Isabella Alexander, *Copyright in Law and the Public Interest in the Nineteenth Century*, Oxford, 2010, 68.

hearing.⁷⁵ It is worth stressing that the maintenance of the *ex parte* injunction would have provided patentees with at least one major advantage, at least until the defendants were able to enter a satisfactory answer. Other evidence suggests that patentees could wield the injunction or the threat of injunction to intimidate infringers. In 1732, for example, Lancashire textile manufacturers petitioned against a proposed extension to the term of Thomas Lombe's patent for silk-throwing machinery. In their petition they complained of being 'threatened by Sir Thomas Lombe with prosecutions for so doing under pretence of infringing upon his sole right of using and enjoying his invention; and have thereupon been obliged to desist'.⁷⁶ There are, however, other occasions where a patentee's bluff was eventually called. Richard Newsham, for example, held a patent for an early steam engine, and by threatening legal action, he was able to cajole his competitor John Ffowke into an agreement for four years. The arrangement, however, ended acrimoniously and Newsham was obliged to bring a Chancery suit against Ffowke in 1729.⁷⁷

Intriguingly, in all three cases where a hearing was held before the Lord Chancellor, the same order was made. The patentees were awarded a temporary injunction lasting twelve months but, to obtain a permanent injunction, they were ordered to try their right in the common law courts.⁷⁸ In the hearing for *Kay v Mills*, for example, it was stated that both sides were 'to be bound by the event of such tryall of such action as to the validity of the said letters patent'.⁷⁹ Where the right of the plaintiff was disputed, ordering a trial at common law was also standard practice in copyright cases.⁸⁰ At least one of these cases was actually tried at common law, *Newsham v Gray* (the same Newsham who was embroiled in the dispute with Ffowke). After reciting the order for the plaintiff to go to common law, the order book states that, 'in Trinity term 1741 the plaintiff brought his action against the defendant in the King's Bench and upon a long tryal in such action of 7th of July last, the plaintiff was nonnsuited'.⁸¹ Frustratingly, the case does not appear in the index to the King's Bench plea rolls for KB 122, so it is impossible to explain why Newsham lost (attorneys increasingly failed to file the enrolments of cases they were involved with).⁸² Nonetheless, this order demonstrates that the common law courts had jurisdiction over the legality of patents before 1753. This is also consistent with the provisions of the Statute of Monopolies, the second section of which states that:

all such commissions, grants, licences, charters, letters patents, proclamations, inhibitions, restraints, warrants of assistance, and all other matters and things

⁷⁵PRO, C 33/385, fo.422.

⁷⁶Parliamentary Papers, 'The Case of the Manufacturers of Woollen, Linnen, Mohair and Cotton Yarn', 1732 (House of Commons).

⁷⁷MacLeod, *Inventing the Industrial Revolution*, 70; *Newsham v Ffowke* (1729), C 11/699/10.

⁷⁸In *Newsham v Gray* (1737), for example, it was ordered 'that the plaintiff's bill be retained for 12 months and that the plaintiff is to be at liberty to bring an action at law'. PRO, C 33/376, fo.336.

⁷⁹*Kay v Mills* (1747) PRO, C 33/383, fo.698.

⁸⁰Gómez, 'Copyright injunctions', 1233.

⁸¹PRO, C 33/378, fo.190.

⁸²The index for King's Bench Trinity term 1741 is PRO, IND 1/6175.

tending as aforesaid, and the force and validity of them, and every of them, ought to be, and shall be for ever hereafter examined, heard, tried, and determined, by and according to the common laws of this realm, and not otherwise.⁸³

Although the vagaries of common law records from this period militate against the type of exercise that was undertaken here for equity, some judicial comments do indicate a sizeable common law case load relating to patents for invention. For instance, as early as 1684, Justice Withins referred to ‘the cause of new inventions, upon which letters patent actions are brought by every day’s experience’. As MacLeod points out, this is likely to be an exaggeration – but unless Withins was entirely mistaken, there must have been patent cases at common law to prompt him to say this.⁸⁴

IV. CONCLUSION

This article has argued that the structure of patent law before 1753 needs to be entirely revised. The Privy Council did not have an important role in the jurisdiction of patent cases after 1688. Instead, patentees were enforcing their rights in the equitable jurisdiction of the Court of Chancery. Also, although the evidence is sparser, it appears that common law had an important jurisdiction in deciding the validity of patents; in all three cases where a temporary injunction was awarded, none were referred to the Privy Council.

This has important implications for how the patent grant during this period should be conceptualized. In the early modern period, patents can be viewed as ‘privileges’, royal favours awarded on the understanding that the invention would be in the public benefit, however defined.⁸⁵ As such, disputes arising from patents were decided in the monarch’s own tribunal; in the case of post-Restoration England this was the Privy Council, and we have seen how considerations of the public benefit were of primary importance in deciding cases before the Council. With the decline of the Council’s jurisdiction, supplanted by the equity and common law courts, patents began to be adjudicated in terms more akin to the modern intellectual property right. For example, although MacLeod suggests that the ‘benefit of the public’ persisted as a matter for judicial enquiry until the end of the eighteenth century, this does not appear to be the case.⁸⁶ Even before 1750, it was unusual for defendants in equity to claim that a patent was not in the public good, and as such void. That they generally eschewed the ‘public good’ as a line of attack against the patent implies that defendants considered it to be an argument unlikely to succeed.⁸⁷

⁸³21 Jac. I, c.3.

⁸⁴Chief Justice Jeffreys also spoke in 1684 of ‘all the actions that have been brought upon that Statute of Monopolies, by the patentees of new inventions, as there have been multitudes in my Lord Hale’s time, and at all times’, corroborating Withins, and suggesting that there is a grain of truth in both comments. MacLeod, *Inventing the Industrial Revolution*, 68–69.

⁸⁵Mario Biagioli, ‘Patent Republic: Representing Inventions, Constructing Rights and Authors’, 73 *Social Research* (2006), 1130.

⁸⁶MacLeod, *Inventing the Industrial Revolution*, 31.

⁸⁷This is not to claim that defendants *never* attacked the supposed ‘convenience’ of a patent. In *Kay v Tattersall*, for example, the defendants claimed ‘that the use of the shuttle pretended to be found out and invented by the complainant Kay is very hurtfull prejudicial and inconvenient to the country in generall in regard that the pieces or cloths thereby wrought and woven are not so well manufactured nor so good

Instead, it was much more common for them to impugn the alleged novelty of the invention. Later, in *Arkwright v Nightingale* (1785), when defence counsel did attempt to argue that Arkwright's patent was invalid 'by the statute against monopolies [as] no patent is to prevail that is generally inconvenient or against the public trade', Lord Loughborough quashed the argument, declaring, 'we must never decide private rights upon any idea of public benefit. I must tell the jury that they must shut out that part of the argument. I cannot let a cause between A and B be determined upon consequential reasons that it is beneficial to the public that B should prevail'.⁸⁸ Because the law courts did not consider the public interest, patent disputes were decided in a more consistent and transparent manner than in the Privy Council.

This discussion of patent law also has important implications for the potential role of patents in encouraging technological development. In theory, by allowing them to exclude (temporarily) other parties from using their inventions, patents are supposed to help inventors appropriate returns from inventive activities, incentivizing the development of new technology. Previous historians of the patent system (Harold Dutton and Christine MacLeod), have argued that until c.1830, patents were difficult to enforce.⁸⁹ The dearth of reported legal cases was taken as evidence for this, and MacLeod argues that 'a patent may not often have been worth the parchment it was written on'.⁹⁰ Because patents seldom provided the means with which to prevent competitors from adopting new technology, many inventors apparently preferred to work in secret, especially before 1830.⁹¹

This is not the place to enter into a full discussion of this issue.⁹² However, because Chancery did provide the means with which patentees could maintain the exclusive working of the patent, it becomes plausible to argue that patents may have encouraged the development of technology during this period. It is striking that the Chancery cases discussed here involved some of the most important inventors of the early Industrial Revolution, figures such as Abraham Darby I,⁹³ Isaac Wilkinson⁹⁴ and John

or fitt for marketts, and the same by taking away employ from any number of hands, tends to lessen workmens wages and has already been the occasion of and may hereafter introduce common and public tumults and disorders by taking away from many of the weavers and manufactures of such commodities the means of their livelihood and subsistence'. *Kay v Tattersall* (1736) PRO, C 11/2452/15, fo.2.

⁸⁸1 HPC 236.

⁸⁹Harold Dutton, *The Patent System and Inventive Activity during the Industrial Revolution, 1750–1852*, Manchester, 1984, ch.4.

⁹⁰MacLeod, *Inventing the Industrial Revolution*, 69.

⁹¹Christine MacLeod, 'Strategies for Innovation: The Diffusion of New Technology in Nineteenth-century British Industry', 45 *Economic History Review* (1992), 304.

⁹²But see Sean Bottomley, *The British Patent System during the Industrial Revolution: From Privilege to Property, 1700–1852*, Cambridge, forthcoming.

⁹³Although Abraham Darby I is credited with being the first iron-founder who was able to replace charcoal with coke in the manufacture of iron, coal had been used in iron smelting beforehand. Darby's real contribution was to find a commercial application for pig iron in the casting of pots and other ironware goods. This explains why Darby patented his casting process but not iron smelting with coke. Robert Allen, *The Industrial Revolution in Global Perspective*, Cambridge, 2009, 222–223. It was this process of casting which was litigated in *Darby v Bartlett* (1716) PRO, C 11/1721/15.

⁹⁴Isaac Wilkinson was another iron-founder, who had hoped to adopt Darby's coke smelting on a much more ambitious scale, but was to be disappointed in most of his endeavours. Profitable, however, was

Kay.⁹⁵ That these inventors went to the trouble of obtaining patents, and then undertook legal action to uphold them, indicates, at the very least, the importance of patent protection to their commercial endeavours.⁹⁶

his ironing box used in the manufacture of linens, which was litigated in *Wilkinson v Sunderland* (1752) PRO, C 11/2514/9. For a brief discussion on Wilkinson and his ironing box see Chaloner, *People and Industries*, 22.

⁹⁵John Kay invented the flying shuttle which greatly improved labour productivity in weaving, and the patent was the most heavily litigated I have found in this period. There are ten surviving pleadings in C 11 relating to Kay's patent. Kay apparently had a difficult, uncooperative personality and ultimately the legal troubles involved with his patent forced him to move to France in 1747. Joel Mokyr, *The Enlightened Economy: An Economic History of Britain, 1700–1850*, New Haven, CT, 2009, 420. 1747 was also the year when the fourteen-year patent term for the flying shuttle expired.

⁹⁶Mario Biagioli also notes in his study on the patenting of scientific instruments in early modern Europe that from 1700 in England, 'patents became increasingly sought and used as legal tools'. This is in contrast to the situation in France, where patents were primarily used as means of securing prestige and patronage. Mario Biagioli, 'From Print to Patents: Living on Instruments in Early Modern Europe', 44 *History of Science* (2006), 144.